The stationary-start maintenance model and its delay-period predictions

Working note

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Status: superseded, kept as a record. This note was the working document in which the stationary start was worked out and its error bounded. Its conclusions are now in the manuscript's Appendix, in a shorter and slightly different form: the manuscript states the approximation as a remark whose error is *computed* over the sampled domain (1.4×10^{-5}) rather than bounded under the conditions $\lambda \geq 5$, $\kappa \geq 0.2$, $a \leq 5$ that Theorem 2 below assumes, and it drops the lemmas that supported that bound. Where the two differ, the manuscript is current. Nothing here is cited by the manuscript.

Abstract

One change to the maintenance model — initialising the Ornstein–Uhlenbeck maintenance noise at its stationary law rather than at zero — makes the transient negligible over the delay-period analysis window under three conditions that can be argued on their own terms. The rate is then the *stationary* Kac–Rice rate, with an explicit error bar, and every prediction follows in a line. Section 1 states the model and the rate, Section 2 the approximation theorem and its conditions, Section 3 the proof, Section 4 the predictions. All statements are verified numerically (Section 5).

1 The model and the rate

Everything before stimulus offset is unchanged. Maintenance begins at $\tau = t_{\text{offset}} + L_{\text{motion}}$, at which time the gate-facing signal y equals the maintained target m. During maintenance,

$$\mathrm{d}\varepsilon(t) = -\kappa\,\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \qquad \mathrm{d}y(t) = \lambda\,[\,m + \varepsilon(t) - y(t)\,]\,\mathrm{d}t, \tag{1}$$

and $H := y - m$ obeys $\mathrm{d}H = \lambda(\varepsilon - H)\mathrm{d}t$.

The change. $H(\tau) = 0$ is *forced*: the filter output is continuous and m is defined as the value of y at τ. $\varepsilon(\tau)$ is free; the present model sets it to 0. We propose instead

$$\varepsilon(\tau) \sim \mathrm{N}\big(0,\ \sigma_\varepsilon^2/2\kappa\big) \quad \text{independent of } W_\varepsilon, \tag{SS}$$

the stationary law of the first equation in (1).

Why. ε represents fluctuations intrinsic to holding a value; nothing makes such a process sit exactly at its mean at the instant maintenance begins, and no measurement constrains it. The choice matters because it is the *only* source of a slow transient in the model. The covariance of (ε, H) relaxes as a combination of three modes, $e^{-2\kappa\rho}$, $e^{-(\lambda+\kappa)\rho}$ and $e^{-2\lambda\rho}$ (Remark 3); the cold start excites all three and the first can be as slow as $1/\kappa \sim 5$ s, so the analysis window lies inside the transient and every prediction is asymptotic or conditional. Under (SS) the first mode has zero coefficient and relaxation is at least as fast as λ.

Notation. $\rho := t - \tau$; β/δ is the speed of temporal preparation, with $\beta = 0$ under Model 1; $d(t) = g - |m| - (\beta/\delta)(t - t_{\text{offset}})^+$ is the distance to the gate; φ, Φ are the standard normal density and c.d.f., and $g(u) := u\Phi(u) + \varphi(u)$. Since $\varphi'(u) = -u\varphi(u)$, the two middle terms of $g'(u) = \Phi(u) + u\varphi(u) + \varphi'(u)$ cancel, so

$$g' = \Phi, \qquad g'' = \varphi, \tag{2}$$

and g is positive, strictly increasing and strictly convex, with $g(0) = 1/\sqrt{2\pi}$. Write $s^2 := \sigma_\varepsilon^2/2\kappa$ and

$$A_\infty := \frac{\lambda s^2}{\lambda + \kappa}, \qquad B_\infty := \frac{\lambda^2 \sigma_\varepsilon^2}{2(\lambda + \kappa)}, \qquad \omega := \sqrt{B_\infty / A_\infty} = \sqrt{\lambda\kappa}, \qquad a_\infty := \frac{d}{\sqrt{A_\infty}}. \tag{3}$$

These are the stationary variance of H, the stationary variance of $\dot{H}$, the characteristic frequency, and the standardised distance to the gate; at stationarity $\operatorname{Cov}(H, \dot{H}) = 0$.

The rate. Let $Z := s_{\text{resp}} H - d$, so partial activations are upcrossings of zero by Z and $\dot{Z} = \dot{H} + \beta/\delta$. The Kac–Rice formula for a differentiable Gaussian process — which requires no stationarity — gives

$$\nu(t) = p_{Z(t)}(0)\, \mathbb{E}\big[\dot{Z}\mathbf{1}_{\{\dot{Z}>0\}} \mid Z = 0\big] = \frac{\varphi(a)}{\sigma_H}\, G, \qquad G := \mathbb{E}\big[\dot{Z}\mathbf{1}_{\{\dot{Z}>0\}} \mid Z = 0\big], \tag{4}$$

with $a := d/\sigma_H$. Since $(Z, \dot{Z})$ is jointly Gaussian, $\dot{Z}$ given $Z = 0$ is $\mathrm{N}(M, S)$ with

$$M = \frac{\beta}{\delta} + \frac{c_H}{\sigma_H^2}\, d, \qquad S = \sigma_{\dot{H}}^2 - \frac{c_H^2}{\sigma_H^2}, \qquad u := \frac{M}{\sqrt{S}}, \tag{5}$$

and the expected positive part of an $\mathrm{N}(M, S)$ variable is $M\Phi(u) + \sqrt{S}\,\varphi(u)$. Substituting $u = M/\sqrt{S}$ into g gives $\sqrt{S}\, g(u) = \sqrt{S}\,(M/\sqrt{S})\Phi(u) + \sqrt{S}\,\varphi(u)$, so

$$G = M\Phi(u) + \sqrt{S}\,\varphi(u) = \sqrt{S}\, g(u). \tag{6}$$

Writing G this way is what makes the estimates of Section 3 work: it separates the scale $\sqrt{S}$ from the shape $g(u)$, and g has the two properties used there — $g' = \Phi \le 1$ and $\log g$ concave.

Proposition 1 (the stationary rate)**.** *Write ν_{stat} for (4) evaluated at the stationary moments (3), that is with $\sigma_H^2 = A_\infty$, $\sigma_{\dot{H}}^2 = B_\infty$, $c_H = 0$. Then*

$$\nu_{\text{stat}} = \varphi(a)\left[\frac{\beta}{\delta\sigma_H}\Phi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right) + \omega\,\varphi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right)\right], \tag{7}$$

whose bracket does not depend on d. Consequently:

(i) ***Model 2*** *($\beta > 0$): $a(\rho) = a(0) - \beta\rho/(\delta\sigma_H)$ is affine, so exactly*

$$\frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} = \frac{\beta}{\delta\sigma_H}\, a(\rho) > 0, \qquad \frac{\mathrm{d}^2\ln\nu_{\text{stat}}}{\mathrm{d}\rho^2} = -\left(\frac{\beta}{\delta\sigma_H}\right)^2. \tag{8}$$

(ii) ***Model 1*** *($\beta = 0$): $M = 0$, $u = 0$, a is constant, and*

$$\nu_{\text{stat}} = \frac{\omega}{2\pi}\, e^{-a^2/2}, \qquad \frac{\mathrm{d}\ln\nu_{\text{stat}}}{\mathrm{d}\rho} \equiv 0. \tag{9}$$

(iii) Under both models $\nu_{\text{stat}} = \varphi(a) \times$ (a factor independent of d), so ν_{stat} is strictly decreasing in a, with no condition.

Proof. Setting $c_H = 0$ in (4) gives (7) and $\ln\nu_{\text{stat}} = \text{const} - a^2/2$, whence (8) since a is affine, and (iii). For (ii), $\beta = 0$ gives $M = 0$, $g(0) = 1/\sqrt{2\pi}$, $G = \sqrt{B_\infty}/\sqrt{2\pi} = \omega\sigma_H/\sqrt{2\pi}$, so $\nu_{\text{stat}} = \varphi(a)\omega/\sqrt{2\pi} = \omega e^{-a^2/2}/2\pi$, constant in ρ. □

2 The stationary approximation

Theorem 2. *Assume*

$$\text{(i) } \lambda \geq 5\,\mathrm{s}^{-1}, \qquad \text{(ii) } \kappa \geq 0.2\,\mathrm{s}^{-1}, \qquad \text{(iii) } a_\infty \leq 5. \tag{10}$$

Then throughout the analysis window $\rho \in [1,3]$ s, and for both models:

(a) $\nu(\rho) = \nu_{\mathrm{stat}}(\rho)\, e^{\theta(\rho)}$ *with* $|\theta(\rho)| \leq \mathcal{E}(\rho)$, *where* $\mathcal{E}(1) \leq 0.36$ *and* $\mathcal{E}$ *decays like* $e^{-\lambda\rho}$: *the rate is exact to within a factor* 1.43 *at the start of the window and* 1.0004 *at its end;*

(b) *the least-squares slope* c_1 *and quadratic coefficient* c_2 *of* $\ln\nu$ *on the* 20 *bins of the window satisfy*

$$\left|c_1 - c_1^{\mathrm{stat}}\right| \leq 0.081\ \mathrm{s}^{-1}, \qquad \left|c_2 - c_2^{\mathrm{stat}}\right| \leq 0.145\ \mathrm{s}^{-2};$$

(c) *under* $\lambda \geq 6$ *these become* $0.027\,\mathrm{s}^{-1}$ *and* $0.049\,\mathrm{s}^{-2}$; *under* $\lambda \geq 8$, 0.0030 *and* 0.0058.

The observed log-rate slope is $+0.867\,\mathrm{s}^{-1}$, so (b) is a 9% error bar on it and (c) a 3% one. *From Section 4 on we assume* (10), *work with* ν_{stat}, *and carry (b) as the error bar.*

2.1 The three conditions

Each is argued without reference to the measured rate.

(i) $\lambda \geq 5\,\mathrm{s}^{-1}$**, i.e.** $1/\lambda \leq 200$ **ms.** $1/\lambda$ is the time constant of the low-pass filter carrying motor preparation to the gate. Dendauw et al. estimate $\lambda \approx 20\,\mathrm{s}^{-1}$ (50 ms) in an immediate-report task, so (10) is four times more permissive than the published value. λ is in principle identifiable from these data, since it sets how fast the rate follows the stimulus; that check has not been carried out.

(ii) $\kappa \geq 0.2\,\mathrm{s}^{-1}$**, i.e.** $1/\kappa \leq 5$ **s.** $1/\kappa$ is the correlation time of the maintenance noise: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so it is the lag at which the fluctuation has decorrelated to $1/e$. Requiring it to be at most one trial says that what is maintained fluctuates *within* a trial; a longer correlation time describes a drift carried across trials, for which the model has no term and which the randomised design averages out rather than expresses as a delay-period ramp. (i) and (ii) are substitutes: $\lambda \to \lambda + 1$ buys about what $\kappa \to 2\kappa$ buys, so only one needs to be argued hard.

(iii) $a_\infty \leq 5$**.** Not fitted. By (9) the expected number of partial activations per 2 s delay window is $(\omega/\pi)e^{-a^2/2}$; at the largest ω in play that is one per 19 trials at $a = 3.3$, one per 246 at $a = 4$, and *one per* 22 000 *at* $a = 5$. So (iii) says only that the model must produce partial activations at all — the experiment comprises of order 10^4 trials — not that it must reproduce the observed rate. Inverting (9) at the observed rates would give the sharper $a \leq 3.3$, tightening every bound by about a third; it is not used.

3 Proof of Theorem 2

The proof runs through four lemmas: the exact moments, a bound on how far they are from stationary, the resulting pointwise error on $\ln\nu$, and its transfer to the fitted coefficients. Nothing in this section is used elsewhere.

Lemma 3.1 (exact moments). Under (1) and (ss) with $H(\tau) = 0$, put $E_1 := e^{-(\lambda+\kappa)\rho}$, $E_2 := e^{-2\lambda\rho}$ and

$$X(\rho) := \frac{(\lambda+\kappa)(E_2 - E_1)}{\lambda - \kappa} \qquad \big(= -2\lambda\rho\, E_2 \ \text{ when } \lambda = \kappa\big). \tag{11}$$

Then for every $\rho > 0$

$$\frac{\sigma_H^2}{A_\infty} = 1 + X - E_1, \qquad \frac{c_H}{A_\infty} = -\lambda X, \qquad \frac{\sigma_{\dot H}^2}{B_\infty} = 1 + \frac{\lambda}{\kappa}\big(X + E_1\big). \tag{12}$$

Proof. (ε, H) is a linear diffusion with $F = \left(\begin{smallmatrix} -\kappa & 0 \\ \lambda & -\lambda \end{smallmatrix}\right)$ and $Q = \operatorname{diag}(\sigma_\varepsilon^2, 0)$, so its covariance solves $\dot P = FP + PF^\top + Q$ with $P(0) = \operatorname{diag}(s^2, 0)$. The system is triangular:

$$\dot P_{\varepsilon\varepsilon} = -2\kappa P_{\varepsilon\varepsilon} + \sigma_\varepsilon^2, \qquad \dot P_{\varepsilon H} = -(\kappa+\lambda)P_{\varepsilon H} + \lambda P_{\varepsilon\varepsilon}, \qquad \dot P_{HH} = 2\lambda(P_{\varepsilon H} - P_{HH}),$$

all starting from $(s^2, 0, 0)$. Since $\dot P_{\varepsilon\varepsilon}(0) = -2\kappa s^2 + \sigma_\varepsilon^2 = 0$, the stationary start freezes the first component: $P_{\varepsilon\varepsilon} \equiv s^2$. Then $P_{\varepsilon H} = A_\infty(1 - E_1)$ and

$$P_{HH} = 2\lambda \int_0^\rho e^{-2\lambda(\rho - v)} P_{\varepsilon H}(v)\, \mathrm{d}v = A_\infty \left[1 - E_2 - 2\lambda \frac{E_1 - E_2}{\lambda - \kappa}\right],$$

and writing $-2\lambda = -(\lambda+\kappa) - (\lambda - \kappa)$ in the numerator gives $P_{HH}/A_\infty - 1 = X - E_1$. Since $\dot H = \lambda(\varepsilon - H)$, $c_H = \lambda(P_{\varepsilon H} - P_{HH}) = -\lambda A_\infty X$, and $\sigma_{\dot H}^2 = \lambda^2(P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}) = B_\infty + \lambda^2 A_\infty(X + E_1)$ using $\lambda^2(s^2 - A_\infty) = B_\infty$ and $\lambda^2 A_\infty / B_\infty = \lambda/\kappa$. The $\lambda = \kappa$ form follows by l'Hôpital. □

Remark 3 (the three modes). The operator $P \mapsto FP + PF^\top$ has eigenvalues $\{-2\kappa, -(\lambda+\kappa), -2\lambda\}$ — the pairwise sums of the eigenvalues of F — so a general initial condition relaxes as a combination of $e^{-2\kappa\rho}$ (the ε–ε block), $e^{-(\lambda+\kappa)\rho}$ (the cross term) and $e^{-2\lambda\rho}$ (the H–H block). Under (ss) the first has zero coefficient, because $P_{\varepsilon\varepsilon}$ is constant, and (12) contains only E_1 and E_2. The slower of those two is $e^{-\mu\rho}$ with $\mu := \lambda + \min(\lambda, \kappa) \geq \lambda$; it equals $e^{-2\lambda\rho}$ only when $\kappa \geq \lambda$. Relaxation is therefore at least as fast as λ, uniformly in κ.

Lemma 3.2 (distance from stationarity). With $\mu := \lambda + \min(\lambda, \kappa)$ and

$$\mathcal{D} := \min\left\{\rho\, e^{-\mu\rho},\ \frac{E_1 + E_2}{|\lambda - \kappa|}\right\} \quad \text{(first entry only when } \lambda = \kappa\text{)}, \tag{13}$$

put

$$\bar\Delta := (\lambda+\kappa)\mathcal{D} + E_1, \quad \bar\gamma := \lambda(\lambda+\kappa)\mathcal{D}, \quad \bar e := \min\left\{\frac{\lambda}{\kappa}\bar\Delta,\ \frac{\lambda\big[(\lambda+\kappa)E_2 + 2\kappa E_1\big]}{\kappa|\lambda - \kappa|}\right\}. \tag{14}$$

Then $\big|\sigma_H^2/A_\infty - 1\big| \leq \bar\Delta$, $0 \leq c_H/A_\infty \leq \bar\gamma$ and $\big|\sigma_{\dot H}^2/B_\infty - 1\big| \leq \bar e$, all decreasing on $\rho \geq 1$ when $\lambda \geq 1$.

Proof. Everything reduces to bounding $|E_1 - E_2|/|\lambda - \kappa|$, since by Lemma 3.1 $X = (\lambda + \kappa)(E_2 - E_1)/(\lambda - \kappa)$ and, adding and subtracting $(\lambda+\kappa)E_1$ in the numerator, $\sigma_{\dot H}^2/B_\infty - 1 = (\lambda/\kappa)\big[(\lambda+\kappa)E_2 - 2\kappa E_1\big]/(\lambda - \kappa)$. Two bounds are available and we take the smaller. *(a)* By the mean value theorem applied to $x \mapsto e^{-x\rho}$ on the interval with endpoints 2λ and $\lambda + \kappa$ — whose length is $|\lambda - \kappa|$ and whose left end is μ — $|E_1 - E_2|/|\lambda - \kappa| \leq \rho e^{-\mu\rho}$. *(b)* Trivially $|E_1 - E_2| \leq E_1 + E_2$. Entry (a) is sharp when $\lambda \approx \kappa$, where (b) is singular; (b) is sharp when the rates are far apart, where (a)'s worst-case exponent μ is pessimistic. The sign $X \leq 0$ follows by inspection of (11), giving $c_H \geq 0$. The second entry of $\bar e$ comes from bounding the two exponentials separately. For monotonicity, $\frac{\mathrm{d}}{\mathrm{d}\rho}(1 + c\rho)e^{-\mu\rho} = e^{-\mu\rho}(c - \mu - \mu c\rho)$ with $c = \lambda + \kappa \geq \mu$ is negative for $\rho > (c - \mu)/(\mu c)$, and $(c - \mu)/(\mu c) \leq 1/(2\lambda) \leq 1/2$ when $\lambda \geq 1$; the remaining entries are sums of decaying exponentials. □

Lemma 3.3 (pointwise error). Fix $\rho \geq 1$, evaluate $\bar{\Delta}, \bar{\gamma}, \bar{e}$ at ρ, and write $u_s := \beta/(\delta\sqrt{B_\infty})$, $c_g := \Phi(0)/g(0) = 1.2533$ and $\eta := \bar{e} + \bar{\gamma}^2/\big(\omega^2(1-\bar{\Delta})\big)$. If $\bar{\Delta} < 1$ and $\eta < 1$ then, for both models,

$$\big|\ln \nu(\rho) - \ln \nu_{\mathrm{stat}}(\rho)\big| \leq \mathcal{E}(\rho) := \frac{(a_\infty^2+1)\bar{\Delta}}{2(1-\bar{\Delta})} + \frac{\eta}{2(1-\eta)} + c_g\left[\frac{\bar{\gamma}\, a_\infty}{\omega(1-\bar{\Delta})\sqrt{1-\eta}} + u_s\left(\frac{1}{\sqrt{1-\eta}} - 1\right)\right]. \tag{15}$$

Proof. Write $\Delta := \sigma_H^2/A_\infty - 1$, $\gamma := c_H/A_\infty$, $E := \sigma_{\dot{H}}^2/B_\infty - 1$, bounded by Lemma 3.2. From (4),

$$\ln \nu - \ln \nu_{\mathrm{stat}} = -\tfrac{1}{2}\big(a^2 - a_\infty^2\big) - \tfrac{1}{2}\ln\frac{\sigma_H^2}{A_\infty} + \ln\frac{G}{G_s}, \qquad G = \sqrt{S}\, g(u), \quad G_s = \sqrt{B_\infty}\, g(u_s).$$

Throughout, $|\ln(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$.

(1) $a^2 = a_\infty^2/(1+\Delta)$, so $|a^2 - a_\infty^2| \leq a_\infty^2\bar{\Delta}/(1-\bar{\Delta})$; halving and adding $\frac{1}{2}|\ln(1+\Delta)| \leq \bar{\Delta}/(2(1-\bar{\Delta}))$ gives the first term.

(2) $S = B_\infty\big[(1+E) - \gamma^2/(\omega^2(1+\Delta))\big]$ since $A_\infty/B_\infty = \omega^{-2}$, so $|S/B_\infty - 1| \leq \eta$, hence $S \geq B_\infty(1-\eta) > 0$ and $\frac{1}{2}|\ln(S/B_\infty)| \leq \eta/(2(1-\eta))$: the second term.

(3) $(\ln g)' = \Phi/g$ and $(\ln g)'' = -D/g^2 < 0$, where $D := \Phi^2 - \varphi g$ satisfies $D' = \varphi(\Phi + ug) > 0$ and $D(-\infty) = 0$, so $D > 0$; thus Φ/g is decreasing and $\sup_{u\geq 0}(\ln g)' = \Phi(0)/g(0) = c_g$. Since $M = \beta/\delta + \gamma d/(1+\Delta) \geq \beta/\delta \geq 0$ we have $u, u_s \geq 0$, so $|\ln g(u) - \ln g(u_s)| \leq c_g|u - u_s|$. Splitting $u - u_s = \frac{M-\beta/\delta}{\sqrt{S}} + \frac{\beta}{\delta}\big(\frac{1}{\sqrt{S}} - \frac{1}{\sqrt{B_\infty}}\big)$: the first piece is at most $\bar{\gamma}a_\infty/\big(\omega(1-\bar{\Delta})\sqrt{1-\eta}\big)$, using $M - \beta/\delta = \gamma d/(1+\Delta)$, $d = a_\infty\sqrt{A_\infty}$ and $\sqrt{S} \geq \sqrt{B_\infty(1-\eta)}$; the second equals $u_s|(1+x)^{-1/2} - 1|$ with $|x| \leq \eta$, hence at most $u_s(1/\sqrt{1-\eta} - 1)$. Multiplying by c_g gives the third term. □

Lemma 3.4 (transfer to the fitted coefficients). On the 20 bin centres $\rho_1, \ldots, \rho_{20}$ of the window, the least-squares slope and quadratic coefficient are linear functionals $c_j = \sum_i w_i^{(j)} f_i$ of the profile, with $\sum_i |w_i^{(1)}| = 1.504$ and $\sum_i |w_i^{(2)}| = 2.916$. Hence

$$\big|c_j - c_j^{\mathrm{stat}}\big| \leq \sum_i \big|w_i^{(j)}\big|\, \mathcal{E}(\rho_i) \;\leq\; \Big(\textstyle\sum_i \big|w_i^{(j)}\big|\Big)\mathcal{E}(1). \tag{16}$$

Proof. Immediate from Lemma 3.3 and $\mathcal{E}$ decreasing (Lemma 3.2). □

Proof of Theorem 2. Under (10), $\mathcal{E}$ of (15) is increasing in a_∞ and, at $\rho = 1$, maximised over $\kappa \geq 0.2$ at $\kappa = 0.2$; evaluating gives $\mathcal{E}(1) = 0.353$ at $\lambda = 5$, 0.133 at $\lambda = 6$, 0.019 at $\lambda = 8$, which is (a). Feeding $\mathcal{E}(\rho_i)$ into the first form of (16) — which is 7 to 10 times smaller than the uniform second form, since $\mathcal{E}$ decays like $e^{-\mu\rho}$ across the window — gives (b) and (c):

$\lambda \geq$	$1/\lambda$	$\mathcal{E}(1)$	slope error	curvature error
4	250 ms	0.96	0.253	0.446
5	200 ms	0.353	0.081	0.145
6	167 ms	0.133	0.027	0.049
8	125 ms	0.019	0.0030	0.0058

□

Remark 4. Lemma 3.2 bounds the *moments*; the observables are a slope and a curvature, that is, *derivatives*, and a bound on a function does not bound its derivative. Lemma 3.3 bridges the gap by bounding $\ln \nu$ itself, pointwise, after which Lemma 3.4 is elementary.

4 Predictions

We assume (10) throughout and work with ν_{stat}, carrying Theorem 2(b) as the error bar. The delay-period observations are: **(L1)** the level rises with stimulus duration; **(L2)** the level rises with motion coherence; **(L3)** the coherence effect on the level is *larger* after the long stimulus; **(S)** the log-rate slope is $+0.867\,\text{s}^{-1}$.

Corollary 5 (L1, L2)**.** From Proposition 1(iii), plus the saturation argument of the main text, which concerns the distribution of m and is untouched by (ss). *Duration and coherence each raise the fraction of trials whose maintained target has reached the cap θ, hence lower d in the sense of first-order stochastic dominance; since ν_{stat} is strictly decreasing in d, and averaging over trials preserves the ordering, both raise the level. Predicted by* both *models:* not discriminating.

Corollary 6 (L3)**.** From Proposition 1(iii) and the same saturation argument. *As the capped fraction tends to one the coherence levels converge, and duration raises that fraction at every coherence, so the coherence effect on the level must* shrink *with duration. Predicted by both models and* contradicted *by the data: not discriminating, and a genuine disagreement.*

Corollary 7 (S)**.** From Proposition 1 and Theorem 2(b). *Model 1 predicts a log-rate slope of exactly* 0 *(Proposition 1(ii)), hence a measured slope of at most* $0.081\,\text{s}^{-1}$ *under* (10)*, and* 0.027 *if* $\lambda \geq 6$*. Model 2 predicts* $\beta a(\rho)/(\delta\sigma_H) > 0$ *(Proposition 1(i)), its only source of a ramp.* Discriminating: the observed $+0.867\,\text{s}^{-1}$ exceeds Model 1's ceiling by a factor of 10.8, or 33 if $\lambda \geq 6$.

Remark 8 (what is no longer claimed)*.* Two delay-period statements that appear in the current draft are dropped here. The *shape* of the ramp no longer discriminates: under (ss) there is no variance filling to decay, both models predict a near-zero curvature (Model 1 exactly 0; Model 2 $-\frac{1}{2}(\beta/\delta\sigma_H)^2 \geq -0.034$ at the observed slope), and the observed -0.0405 ± 0.0416 is consistent with both. The dependence of the *slope on the conditions* is also dropped: it discriminates only at $\lambda \geq 6$, it was absent from the first version of the paper, and it is the only statement requiring the unbinned point-process analysis. What remains is one discriminating prediction, exactly stated, with an explicit error bar.

5 Numerical verification

Claim	Test	Result
Lemma 3.1	vs. numerical integration of $\dot{P} = FP + PF^\top + Q$	max rel. err. 8.9×10^{-12}
Lemma 3.1	the three identities in terms of X	agree to 1.5×10^{-13}
Lemma 3.2	the three bounds, 50-digit arithmetic	worst ratios 1.000000
Lemma 3.3	71 079 points, both models	0 violations, worst ratio 0.612
Lemma 3.4	the functional norms 1.504, 2.916	computed exactly

Grids: $\lambda \in [0.3, 60]$, $\kappa \in [0.01, 60]$, $\rho \in [0.05, 6]$, $\beta/\delta \in \{0, 0.05, 0.3, 1, 3\}$, $a_\infty \in \{0.5, 1, 2, 3.3, 6, 10\}$. The moment bounds were verified in 50-digit arithmetic because the combination $P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}$ cancels catastrophically in double precision; points where $\mathcal{E}$ falls below double-precision resolution were excluded from the Lemma 3.3 check for the same reason.

6 Consequences for the paper

Adopting (ss) removes from the appendix the convolution kernel K, the integrals $\int K^2$ and $\int K'^2$, the auxiliary $W = K' - K^3/2\sigma_H^2$, the log-concavity lemma, the non-stationary derivative

of $\ln \nu$, and the transient content of the present Corollaries A.2 and A.3, together with the point-process analysis of the delay period. What remains is Proposition 1, Theorem 2 with its conditions, and three corollaries; the proof of the theorem can sit in a subsection or be deferred entirely.

The cost is that theory and simulation must agree, so the 1200-draw parameter sweep, the admissibility screen and every effect size would have to be recomputed under (ss).